

Unbounded Orbit-Series and Reductivity Indices of Finite Quandles

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Abstract

For a finite quandle, let k , j , and i denote the least positive indices of local reductivity, trivialization of orbit series, and reductivity, respectively. We show that j cannot be bounded by a function of k , and i cannot be bounded by a function of j . For every $n \geq 3$, we construct a finite quandle Q_n with $k(Q_n) = 2$ and $j(Q_n) \geq \lfloor \log_3 n \rfloor + 1$. For every $m \geq 2$, we construct a finite quandle P_m with $(k(P_m), j(P_m), i(P_m)) = (2, 2, m)$. The first family is obtained from the group of Noce–Tracey–Traustason using a filtration by label size; the second is obtained from finite quotients of the reduced free group that preserve its exact nilpotency class. All passages from groups to quandles, including the shift between the nilpotency class of the inner automorphism group and the reductivity index, are proved explicitly. This gives negative answers to both parts of Question 5.25 of Bonatto–Crans–Nasybullov–Whitney.

Keywords: finite quandle, local reductivity, orbit series, reduced free group, Engel element.

1 Introduction and main results

The classes $\mathcal{LR}_n$ and $t\mathcal{OS}_n$ were introduced in [1], where their relationship with the classes $\mathcal{R}_n$ was investigated. For finite quandles, these finite-depth conditions are equivalent at the level of the unions of the respective classes [1, Corollary 5.19]. Question 5.25 asks whether orbit-series depth can be bounded in terms of the local reductivity index, and whether the reductivity index can be bounded in terms of orbit-series depth. The numbering of the results in [1] used here follows arXiv version v3, dated August 18, 2020.

For a quandle whose corresponding indices exist, set

$$\begin{aligned} k(Q) &= \min\{n \geq 1 : Q \in \mathcal{LR}_n\}, \\ j(Q) &= \min\{n \geq 1 : Q \in t\mathcal{OS}_n\}, \\ i(Q) &= \min\{n \geq 1 : Q \in \mathcal{R}_n\}. \end{aligned}$$

The classes are defined in Section 2.

Theorem 1.1. *For every integer $n \geq 3$, there is a finite quandle Q_n such that*

$$k(Q_n) = 2, \quad j(Q_n) \geq \lfloor \log_3 n \rfloor + 1.$$

Theorem 1.2. *For every integer $m \geq 2$, there is a finite quandle P_m such that*

$$(k(P_m), j(P_m), i(P_m)) = (2, 2, m).$$

The quandle P_m can be chosen to have exactly m inner orbits, each of which is trivial as a quandle.

Corollary 1.3. *There is no function $f : \mathbb{N}_{>0} \rightarrow \mathbb{N}_{>0}$ such that $j(Q) \leq f(k(Q))$ for every finite quandle with defined indices. Nor is there a function g such that $i(Q) \leq g(j(Q))$ for all such quandles.*

Proof. The first family has $k = 2$ and unbounded j . The second has $j = 2$ and unbounded i . $\square$

Theorems 1.1 and 1.2 are proved in Sections 3 and 4. We use two known group constructions: the group of Noce–Tracey–Traustason [2] and the reduced free group as treated by Darné [3]. The group-theoretic input results are stated separately from the consequences for quandles established here.

2 Definitions and a reductivity criterion

We use the left convention for quandles [1]: a quandle is a nonempty set Q with an operation $\triangleright$ satisfying

$$a \triangleright a = a, \quad L_a : b \mapsto a \triangleright b \text{ is bijective}, \quad a \triangleright (b \triangleright c) = (a \triangleright b) \triangleright (a \triangleright c).$$

The group $\text{Inn}(Q) = \langle L_a : a \in Q \rangle$ is its inner automorphism group. Subquandles are also closed under inverse left translations. The trivial operation is given by $a \triangleright b = b$.

Write

$$T_n(a; c_1, \dots, c_n) = ((a \triangleright c_1) \triangleright \dots) \triangleright c_n$$

for the left-associated expression with n operations. The class $\mathcal{R}_n$ is defined by the identity

$$T_n(a; c_1, \dots, c_n) = T_n(b; c_1, \dots, c_n)$$

for all arguments; the class $\mathcal{LR}_n$ is defined by

$$T_n(a; \underbrace{b, \dots, b}_n) = b.$$

An orbit series is defined recursively by

$$Q_0 = Q, \quad Q_{r+1} = \text{Orb}(x_r, Q_r), \quad x_r \in Q_r.$$

At each step, the inner automorphism group of the current subquandle is used. The class $t\mathcal{OS}_n$ consists of quandles for which every such series reaches a singleton quandle no later than step n . Thus j measures the time to reach a singleton, rather than merely the time to stabilization. All three classes with index 1 coincide with the class of trivial quandles. Moreover, $Q \in t\mathcal{OS}_2$ if and only if each inner orbit of Q is trivial as a quandle; such an orbit may contain many elements.

For groups, we use the conventions

$$[a, b] = a^{-1}b^{-1}ab, \quad c^g = g^{-1}cg, \quad \gamma_1(H) = H, \quad \gamma_{r+1}(H) = [\gamma_r(H), H].$$

The trivial group has class zero. Otherwise, exact class c means $\gamma_{c+1}(H) = 1$, $\gamma_c(H) \neq 1$.

Lemma 2.1. *Let S be a nonempty subset of a group, closed under conjugation by its own elements and their inverses. Then $a \triangleright b = a^{-1}ba$ makes S a quandle, and for $a \in S$*

$$\text{Orb}(a, S) = a^{\langle S \rangle}.$$

Proof. Idempotency and bijectivity are immediate, and

$$(a^{-1}ba)^{-1}(a^{-1}ca)(a^{-1}ba) = a^{-1}b^{-1}cba$$

proves self-distributivity. Left translations and their inverses are conjugations by elements of S and their inverses. They generate the action of $\langle S \rangle$, giving the orbit formula. $\square$

Proposition 2.2. *For every nonempty quandle Q and $n \geq 1$,*

$$Q \in \mathcal{R}_n \iff \gamma_n(\text{Inn}(Q)) = 1.$$

In particular, if $\text{Inn}(Q)$ has exact finite class c , then $i(Q) = c + 1$.

Proof. Set $\mathcal{L}(Q) = \{L_a : a \in Q\}$. Self-distributivity gives

$$L_{a \triangleright b} = L_a L_b L_a^{-1}.$$

Thus $a \mapsto L_a$ is an epimorphism of quandles when $\mathcal{L}(Q)$ is equipped with $s \triangleright t = sts^{-1}$. This direction of conjugation is opposite to the convention in Lemma 2.1.

For $n \geq 1$, the identity $\mathcal{R}_{n+1}$ for Q is equivalent to the identity $\mathcal{R}_n$ for $\mathcal{L}(Q)$: equality of the two expressions after the final operation for every final argument means equality of their left translations. Also, $Q \in \mathcal{R}_1$ if and only if $\mathcal{L}(Q)$ is a singleton. Indeed, if all L_a equal f , then idempotency gives $f(a) = a$ for every a . For the iterates $\mathcal{L}_0(Q) = Q$, $\mathcal{L}_{r+1}(Q) = \mathcal{L}(\mathcal{L}_r(Q))$, we obtain

$$Q \in \mathcal{R}_n \iff |\mathcal{L}_n(Q)| = 1.$$

Let $H = \text{Inn}(Q)$. The set $\mathcal{L}(Q)$ generates H and is preserved by conjugations in H , since $hL_a h^{-1} = L_{h(a)}$. The action of H on this set has image $\text{Inn}(\mathcal{L}(Q))$ and kernel $C_H(\mathcal{L}(Q)) = Z(H)$. Consequently,

$$\text{Inn}(\mathcal{L}(Q)) \cong H/Z(H).$$

For the upper central series $Z_0(H) = 1$, $Z_{r+1}(H)/Z_r(H) = Z(H/Z_r(H))$, induction gives

$$\text{Inn}(\mathcal{L}_r(Q)) \cong H/Z_r(H).$$

The quandle $\mathcal{L}_{n-1}(Q)$ is trivial if and only if $\mathcal{L}_n(Q)$ is a singleton; hence the latter condition is equivalent to $H = Z_{n-1}(H)$.

Finally, $H = Z_c(H)$ is equivalent to $\gamma_{c+1}(H) = 1$. This follows by induction on c , starting with $c = 0$: the condition $\gamma_{c+1}(H) = 1$ is equivalent to $\gamma_c(H) \leq Z(H)$, or $\gamma_c(H/Z(H)) = 1$. Applying induction to $H/Z(H)$ gives $H = Z_c(H)$. Taking $c = n-1$ proves the criterion, including $n = 1$. $\square$

Remark 2.3. The criterion corresponds to [1, Proposition 3.2], but is expressed by vanishing of a term of the series to distinguish a bound on the class from its exact value.

3 First family: unbounded depth

3.1 Group-theoretic input

We use the group $\mathcal{G}$ over $\mathbb{F}_2$ constructed in [2]. Write its generators as

$$X = 1 + \text{ad}(x), \quad U_A = 1 + \text{ad}(u_A), \quad V_A = 1 + \text{ad}(v_A), \quad W_A = 1 + \text{ad}(w_A),$$

where A is a nonempty finite subset of $\mathbb{N}$. The operators and their action are defined in [2, Section 2]. All these generators are involutions. Each of the three families U, V, W generates an elementary abelian group. If $A \cap B = \emptyset$, put $A \sqcup B = A \cup B$; otherwise put $A \sqcup B = \emptyset$. The empty label corresponds to the identity. By [2, Lemma 2.3],

$$\begin{aligned} [U_A, V_B] &= U_{A \sqcup B}, & [V_A, W_B] &= W_{A \sqcup B}, \\ [W_A, U_B] &= V_{A \sqcup B}, & [W_A, X] &= U_A, \end{aligned} \tag{1}$$

and X commutes with all U_A, V_A . By [2, Proposition 2.4], every element has a unique normal form $X^\epsilon uvw$, where $\epsilon \in \{0, 1\}$ and u, v, w belong to the corresponding elementary abelian groups. Finally, the proof of [2, Theorem 2.5] establishes

$$[[X^g, X], X] = 1 \quad (g \in \mathcal{G}). \quad (2)$$

Every U_A is nontrivial. To see this in the original operator construction, choose a singleton B disjoint from A : the operator $\text{ad}(u_A)$ sends w_B to the nonzero $v_{A \cup B}$.

For $n \geq 1$, set

$$G_n = \langle X, U_A, V_A, W_A : \emptyset \neq A \subseteq [n] \rangle \leq \mathcal{G}, \quad [n] = \{1, \dots, n\}.$$

When a word is collected into normal form, new labels arise only through $\sqcup$, and therefore remain subsets of $[n]$. It follows that G_n is finite and

$$|G_n| \leq 2^{1+3(2^n-1)}. \quad (3)$$

Here the labels themselves are restricted, rather than only their sizes.

3.2 The filtration and normal closures

For an integer $s \geq 1$, define the subgroup

$$F_s = \langle X, U_A (|A| \geq s), V_A (|A| \geq 2s), W_A (|A| \geq 3s) \rangle \leq G_n,$$

using only labels in $[n]$. Its elements are precisely the normal forms with

$$u \in \langle U_A : |A| \geq s \rangle, \quad v \in \langle V_A : |A| \geq 2s \rangle, \quad w \in \langle W_A : |A| \geq 3s \rangle.$$

Indeed, the only correction terms in the collection process satisfy

$$\begin{aligned} [X, W_{\geq 3s}] &\subseteq U_{\geq 3s}, & [U_{\geq s}, V_{\geq 2s}] &\subseteq U_{\geq 3s}, \\ [W_{\geq 3s}, U_{\geq s}] &\subseteq V_{\geq 4s}, & [V_{\geq 2s}, W_{\geq 3s}] &\subseteq W_{\geq 5s}. \end{aligned}$$

The threshold does not decrease: disjoint labels give the sum of their sizes, whereas intersecting labels give the identity. Hereafter, $U_{\geq t}$ and the analogous symbols denote subgroups generated by labels of the corresponding size.

Lemma 3.1. *For every $s \geq 1$,*

$$\langle X^{F_s} \rangle = F_{3s}.$$

Moreover, $\langle X^{G_n} \rangle = F_1$.

Proof. First, we check that $F_{3s} \trianglelefteq F_s$. In the table, rows correspond to generators of F_{3s} and columns to generators of F_s ; an entry indicates the layer containing a possible nontrivial commutator:

	X	$U_{\geq s}$	$V_{\geq 2s}$	$W_{\geq 3s}$
X	1	1	1	$U_{\geq 3s}$
$U_{\geq 3s}$	1	1	$U_{\geq 5s}$	$V_{\geq 6s}$
$V_{\geq 6s}$	1	$U_{\geq 7s}$	1	$W_{\geq 9s}$
$W_{\geq 9s}$	$U_{\geq 9s}$	$V_{\geq 10s}$	$W_{\geq 11s}$	1

All entries lie in F_{3s} . Since the generators are involutions, this check ensures invariance under conjugation by them in both directions. The subgroup contains X , so $\langle X^{F_s} \rangle \leq F_{3s}$.

For the reverse inclusion, set $K = \langle X^{F_s} \rangle$. It is normal in F_s and contains X . If $|A| \geq 3s$, then $W_A \in F_s$, so $[W_A, X] = U_A \in K$. Every label T of size at least $6s$ can be partitioned as $A \sqcup B$ with $|A|, |B| \geq 3s$; then $[W_A, U_B] = V_T \in K$. For $|T| \geq 9s$, take a partition with

$|A| \geq 6s$, $|B| \geq 3s$. The previous step gives $V_A \in K$, and hence $[V_A, W_B] = W_T \in K$. These are all the generators of F_{3s} .

For the initial step, we check that $F_1 \trianglelefteq G_n$. Possible nontrivial commutators between generators lie in the layers

$$\begin{aligned} [X, W_{\geq 1}] &\subseteq U_{\geq 1}, & [U_{\geq 1}, V_{\geq 1}] &\subseteq U_{\geq 2}, \\ [U_{\geq 1}, W_{\geq 1}] &\subseteq V_{\geq 2}, & [V_{\geq 2}, W_{\geq 1}] &\subseteq W_{\geq 3}, \\ [W_{\geq 3}, U_{\geq 1}] &\subseteq V_{\geq 4}, & [W_{\geq 3}, V_{\geq 1}] &\subseteq W_{\geq 4}, \\ [W_{\geq 3}, X] &\subseteq U_{\geq 3}. \end{aligned}$$

They all lie in F_1 ; the others are the identity or are obtained by inverting a commutator. Thus $\langle X^{G_n} \rangle \leq F_1$. Conversely, $[W_A, X] = U_A$ puts all U_A in the normal closure; partitions into two nonempty parts and $[W_A, U_B]$ give all V_T with $|T| \geq 2$; partitions with $|A| \geq 2$, $|B| \geq 1$ and $[V_A, W_B]$ give all W_T with $|T| \geq 3$. $\square$

3.3 The quandle and its depth

Proof of Theorem 1.1. Let $C_n = X^{G_n}$, and equip $Q_n = C_n$ with the operation $a \triangleright b = a^{-1}ba$. This is a finite quandle by Lemma 2.1. For $a = X^g$, $b = X^h$, conjugation by h^{-1} sends the pair to $X^{gh^{-1}}, X$. Thus (2) gives $[[a, b], b] = 1$ for all $a, b \in C_n$. Interchanging a, b gives $[[b, a], a] = 1$. Since $[b, a] = [a, b]^{-1}$, the commutator $[a, b]$ centralizes both generators. The subgroup $\langle a, b \rangle$ has class at most two. In particular, $a^{-1}ba$ commutes with b , so

$$(a \triangleright b) \triangleright b = b.$$

Hence $Q_n \in \mathcal{LR}_2$.

For $n \geq 3$, choose $|A| = 3$. Then $W_A \in F_1 = \langle C_n \rangle$ and $[W_A, X] = U_A \neq 1$. By Lemma 2.1, the orbit of X in Q_n is nontrivial, so Q_n is not trivial and $k(Q_n) = 2$. The index $j(Q_n)$ exists: by [1, Corollary 5.19], a finite quandle in $\mathcal{LR}_2$ belongs to $t\mathcal{OS}$, meaning that every orbit series reaches a singleton quandle. The length of such a series in a finite quandle is bounded: all inclusions before the singleton term are strict, since equality of consecutive terms would mean that the current term is connected and cannot decrease further. Therefore $Q_n \in \bigcup_{t \geq 1} t\mathcal{OS}_t$.

Choosing the anchor X at every step, set

$$S_0 = C_n, \quad N_r = \langle S_r \rangle, \quad S_{r+1} = X^{N_r}.$$

This is an orbit series by Lemma 2.1. Lemma 3.1 gives

$$N_r = F_{3^r} \quad (r \geq 0), \quad S_r = X^{F_{3^{r-1}}} \quad (r \geq 1).$$

If $r \geq 1$ and $3^r \leq n$, choose $|A| = 3^r$. Then $W_A \in F_{3^{r-1}}$, so $X, X^{W_A} \in S_r$; these elements are distinct because $[W_A, X] = U_A \neq 1$. Consequently, S_r is not a singleton. For $r = \lfloor \log_3 n \rfloor$, the series does not reach a singleton at any of the steps $0, \dots, r$, and $j(Q_n) \geq r + 1$. $\square$

Remark 3.2. Theorem 1.1 establishes a lower bound. Exact values of $j(Q_n)$ and minimal orders of quandles with prescribed indices are not determined here.

4 Second family: unbounded reductivity

4.1 The reduced free group and a finite quotient

Let $F(x_1, \dots, x_m)$ be the free group. Define the reduced free group

$$RF_m = F(x_1, \dots, x_m) / \langle\langle [x_r, x_r^w] : 1 \leq r \leq m, w \in F(x_1, \dots, x_m) \rangle\rangle.$$

Each x_r commutes with all its conjugates. By [3, Propositions 1.3 and 1.18],

$$\gamma_{m+1}(RF_m) = 1, \quad \gamma_m(RF_m) = Z(RF_m) \cong \mathbb{Z}^{(m-1)!}.$$

Here and below, the numbering in [3] follows arXiv version v2. Thus RF_m has exact class m . Choose $1 \neq z_m \in \gamma_m(RF_m)$.

We justify the existence of a finite quotient in which z_m survives. Consider the ring

$$\mathcal{A}_m = \mathbb{Z}\langle X_1, \dots, X_m \rangle / I,$$

where I is generated by words with at least one repeated letter. Its additive basis consists of the empty word and all words without repetitions. For the ideal $J = (X_1, \dots, X_m)$, we have $J^{m+1} = 0$; thus $1 + J$ is a group, with inverses

$$(1 + u)^{-1} = \sum_{t=0}^m (-u)^t.$$

By [3, Corollary 1.13], the Magnus map

$$\mu : RF_m \hookrightarrow 1 + J, \quad x_r \mapsto 1 + X_r$$

is injective. The expansion $\mu(z_m) - 1 = \sum_w c_w w$ has a nonzero integer coefficient c_w . Choose a prime p not dividing this coefficient and reduce the ring modulo p . The group $1 + \bar{J}$ is finite, since $\bar{J}$ has a finite basis over $\mathbb{F}_p$; its order is p^{d_m} , where

$$d_m = \sum_{t=1}^m \frac{m!}{(m-t)!}.$$

Let A_m be the image of RF_m in this group, and let $\phi_m : RF_m \twoheadrightarrow A_m$ be the epimorphism onto the image. Then $\phi_m(z_m) \neq 1$. Epimorphisms preserve the lower central series: $\phi_m(\gamma_t(RF_m)) = \gamma_t(A_m)$, as follows by induction from preservation of commutators. Hence A_m has exact class m .

To make the conjugacy classes of the chosen generators distinct, set

$$\alpha : RF_m \rightarrow (\mathbb{Z}/2\mathbb{Z})^m, \quad x_r \mapsto e_r, \quad G_m = \text{im}(\phi_m, \alpha) \leq A_m \times (\mathbb{Z}/2\mathbb{Z})^m.$$

The map α is well-defined because all the relations lie in the commutator subgroup of the free group. Denote the epimorphism onto G_m by ψ_m , and the images of x_r by a_r . The group G_m is finite, is generated by $a_1, \dots, a_m$, and has exact class m : it is a quotient of RF_m , and its first coordinate preserves z_m . The abelian coordinates e_r are distinct, so a_r, a_s cannot be conjugate for $r \neq s$.

4.2 Orbits and the three exact indices

Proof of Theorem 1.2. Fix $m \geq 2$, and write $G = G_m$ for brevity. Set

$$C_r = a_r^G, \quad P_m = \bigcup_{r=1}^m C_r, \quad u \triangleright v = u^{-1}vu.$$

By Lemma 2.1, this is a finite quandle. The classes C_r are pairwise disjoint. The relations of the reduced free group give $[a_r, a_r^g] = 1$ for every $g \in G$. Any two elements a_r^g, a_r^h commute: conjugation by g^{-1} sends them to $a_r, a_r^{hg^{-1}}$. Thus every C_r is trivial as a quandle.

Consider the homomorphism

$$\rho : G \rightarrow \text{Sym}(P_m), \quad \rho(g)(v) = gv g^{-1}.$$

With the usual composition of permutations, this is a homomorphism; the maps $v \mapsto g^{-1}vg$ form an antihomomorphism. We have $L_u = \rho(u^{-1})$. Since P_m contains all the a_r , it generates G ; hence $\text{Inn}(P_m) = \rho(G)$. Its orbits are exactly $C_1, \dots, C_m$. Every orbit series reaches one of the C_r at its first step and a singleton quandle at its second. Thus $P_m \in t\mathcal{OS}_2$. The quandle is not trivial: otherwise all the a_r would commute pairwise, and the group G that they generate would be abelian, contrary to its class $m \geq 2$. Therefore $j(P_m) = 2$.

The element $u \triangleright v$ lies in the same C_r as v , and therefore commutes with v . It follows that $(u \triangleright v) \triangleright v = v$, so $P_m \in \mathcal{LR}_2$. Nontriviality gives $k(P_m) = 2$.

The kernel of ρ is $C_G(P_m) = Z(G)$, since P_m generates G . Consequently,

$$\text{Inn}(P_m) \cong G/Z(G).$$

The quotient $G/Z(G)$ has exact class $m - 1$. Indeed, $\gamma_m(G) \leq Z(G)$, so $\gamma_m(G/Z(G)) = 1$. If $\gamma_{m-1}(G/Z(G)) = 1$, then $\gamma_{m-1}(G) \leq Z(G)$, forcing $\gamma_m(G) = 1$, a contradiction. Proposition 2.2 now gives $i(P_m) = m$. All the required equalities follow. $\square$

Remark 4.1. The abelian factor $(\mathbb{Z}/2\mathbb{Z})^m$ is needed to ensure exactly m distinct orbits. For unboundedness of i with $j = 2$, A_m itself is sufficient. The prime p is chosen after choosing z_m ; we do not assert that an arbitrary z_m survives for any prime p fixed in advance.

4.3 An example for $m = 2$

To illustrate the index shift, one may take $G = UT_3(\mathbb{F}_2)$ with generators

$$a = 1 + E_{12}, \quad b = 1 + E_{23}, \quad z = 1 + E_{13}.$$

This group has order 8, class 2, and center $\{1, z\}$. The generator classes are $\{a, az\}$ and $\{b, bz\}$. On $P = \{a, az, b, bz\}$, the operation has table

$\triangleright$	a	az	b	bz
a	a	az	bz	b
az	a	az	bz	b
b	az	a	b	bz
bz	az	a	b	bz

There are two trivial orbits of size 2, and $\text{Inn}(P) \cong G/Z(G)$ is nontrivial and abelian. Thus $(k(P), j(P), i(P)) = (2, 2, 2)$.

5 Scope of the results

Both obstructions occur already at the least nontrivial value of the controlling index, namely 2. The first family consists of a single group conjugacy class; the second consists of a union of the classes of chosen generators. It is not the full conjugation quandle on the entire group. Consequently, [1, Corollary 5.24], which concerns the full quandle $\text{Conj}(G)$, does not bound the class of the group in the construction of P_m .

The finite quotient in the second family is chosen to preserve a nonzero commutator of weight m . An arbitrary finite quotient may decrease the class and does not ensure the required index. The proofs establish the existence of these families; optimal orders and the exact depth of the first family are outside the scope of this paper.

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